

Software Architecture Document

Sri Lanka Police Traffic Fine Collection System

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1. Introduction

The purpose of this system is to digitalize traffic fine collection in Sri Lanka by enabling drivers to pay fines through a mobile application or web portal while providing administrative monitoring capabilities to senior police officials.

The solution aims to:

- Reduce manual processing.
- Improve payment efficiency.
- Enable real-time payment verification.
- Provide nationwide monitoring and reporting.
- Support the Sri Lankan government's digital transformation initiative.

2. Functional Requirements

Driver Functions

- Enter Fine Reference Number.
- Enter Fine Category Identifier.
- View fine details.
- Make online payments.
- Receive payment confirmation.

Traffic Police Officer Functions

- Issue traffic fine sheets.
- Receive SMS notification after payment.
- Release driver's license after payment confirmation.

Administrator Functions

- View total collections.
- View district-wise collections.
- View category-wise collections.
- Generate reports and statistics.

3. Non-Functional Requirements

Availability -	System available 24/7
Scalability -	Support nationwide traffic fine processing
Security -	Secure payment processing and data protection
Performance -	Payment confirmation within a few seconds
Reliability -	No loss of transaction data
Maintainability -	Independent deployment of services
Usability -	Mobile-friendly interfaces

4. Assumptions

1. Every traffic fine issued has a unique reference number.
2. Fine details are already available in a central police database.
3. Internet connectivity is available during payment.
4. SMS Gateway service is provided by a third-party vendor.
5. Payment Gateway is provided by a licensed Sri Lankan payment provider.
6. Drivers possess smartphones or internet access.
7. Police officers have registered mobile numbers.

5. Architectural Style: Three-Tier Architecture with REST API

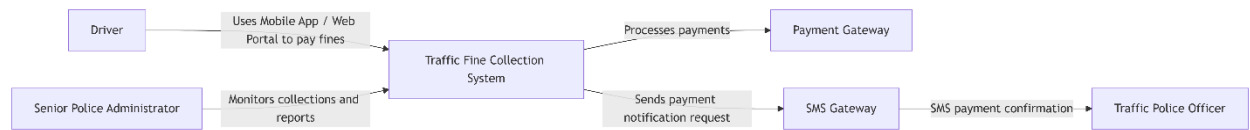
Layers:

1. Presentation Layer
2. Business Layer
3. Data Layer

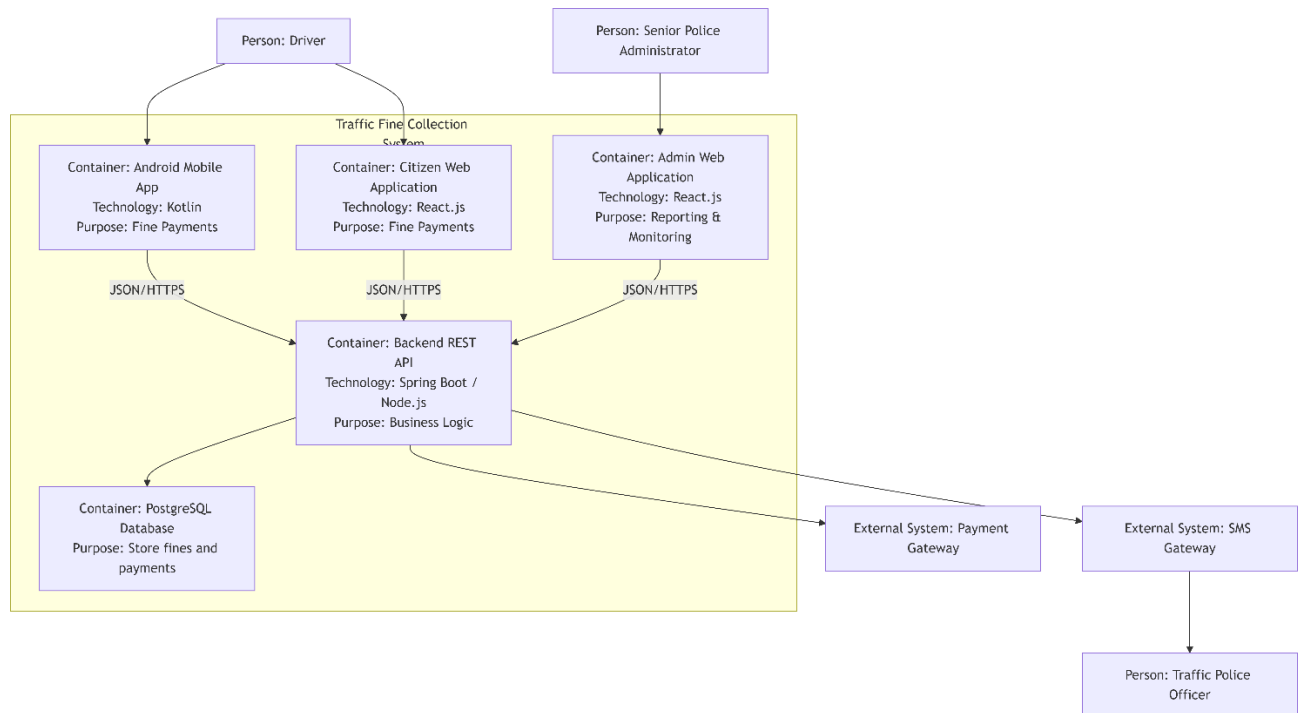
This architecture provides:

- Separation of concerns
- Easier maintenance
- Independent scalability
- Technology flexibility

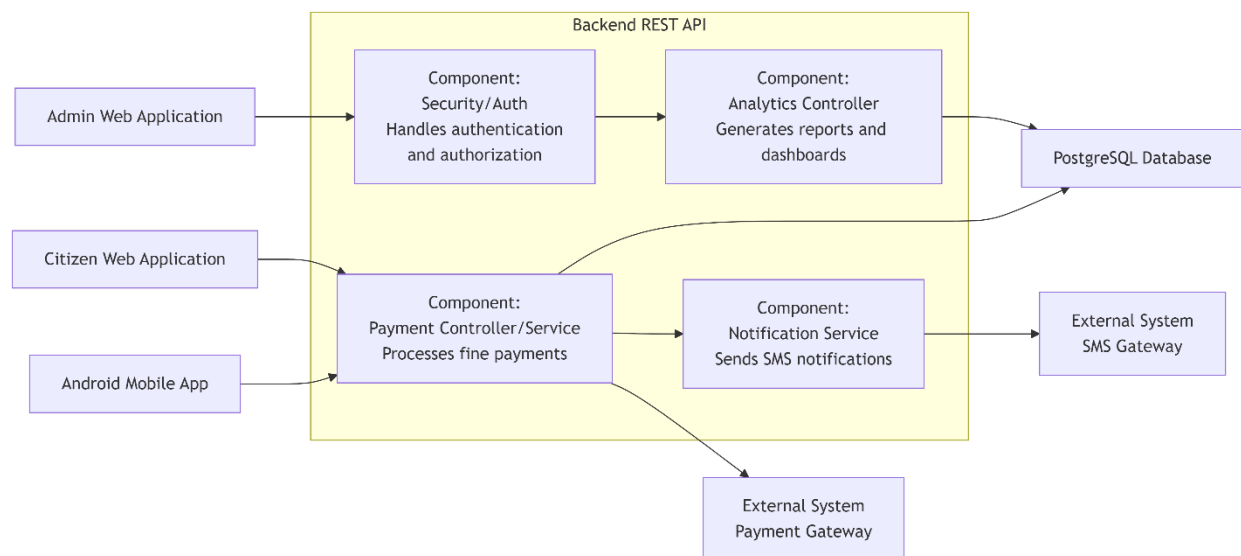
6. C4 Model – Level 1 (System Context Diagram)



7. C4 Model – Level 2 (System Container Diagram)



7. C4 Model – Level 3 (Component Diagram)



8. Technology Selection and Justification

Frontend: React.js

Reasons:

- Fast development
- Reusable components
- Large community support
- Responsive UI

Mobile Application: Kotlin

Reasons:

- Official Android language
- Better performance
- Strong type safety

Backend: Spring Boot

Reasons:

- Enterprise-grade framework
- Excellent REST support
- Security integration
- Easy scalability

Database: PostgreSQL

Reasons:

- ACID compliance
- Reliability
- High performance
- Open source

Authentication: JWT

Reasons:

- Stateless authentication
- Scalable
- Secure API access

9. Security Considerations

HTTPS

All communications encrypted using TLS.

JWT Authentication

Used for:

- Admin users
- System users

Role-Based Access Control (RBAC)

Roles:

- Driver
- Admin

Input Validation

Prevent:

- SQL Injection
- XSS attacks

Secure Payment Processing

Payment data handled only through Payment Gateway.

10. Conclusion

The proposed architecture provides a scalable, secure, and maintainable solution for Sri Lanka's Traffic Fine Collection System. The separation of Mobile App, Public Web Portal, Admin Portal, and Backend REST API ensures modularity and independent deployment. The use of Spring Boot, React, PostgreSQL, JWT authentication, Payment Gateway integration, and SMS notification services enables reliable nationwide digital traffic fine collection and monitoring.